

# Uber Trip Analysis Dashboard

## Microsoft Power BI

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## Introduction

This document explains the complete Power BI development process for the **Uber Trip Analysis Dashboard**.

The project transforms Uber trip-level data into interactive dashboards that help stakeholders understand:

- Booking volume
- Booking value
- Trip distance and duration
- Payment behavior
- Day vs. night demand
- Vehicle performance
- Pickup and drop-off location patterns
- Time-based demand trends
- Detailed trip-level records

The dashboard is designed to support data-driven decisions related to demand planning, driver availability, pricing, vehicle allocation, and operational efficiency.

The documentation follows the same overall approach as the provided sample document: **project overview** → **data preparation** → **data modeling** → **DAX** → **dashboard development** → **business insights** → **recommendations** → **limitations** → **conclusion**.

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## 1. Project Overview

### Project Title

**Uber Trip Analysis Dashboard**

### Objective

The business requirement is to analyze Uber trip data using Power BI to gain insights into:

- Booking trends
- Revenue generation
- Trip efficiency
- Trip distance
- Trip duration
- Location demand

- Vehicle performance
- Time-based booking patterns

The project is intended to help stakeholders make data-driven decisions.

## Key Business Questions

The dashboard addresses questions such as:

1. How many trips were booked?
2. What is the total booking value?
3. What is the average booking value?
4. How much distance was covered?
5. What is the average trip distance?
6. What is the average trip duration?
7. Which payment method is most frequently used?
8. How do day trips compare with night trips?
9. Which vehicle type performs best?
10. Which locations have the highest booking demand?
11. What are the most frequent pickup and drop-off locations?
12. What is the longest trip?
13. During which hours is demand highest?
14. Which days generate the highest booking activity?
15. Which hour/day combinations show the strongest demand?

## 2. Tools & Technologies Used

| Tool / Technology                                                                                                           | Purpose                                |
|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
|  Microsoft Excel                         | Source data                            |
|  Microsoft Power BI Desktop              | Dashboard development                  |
|  Power Query / Power BI Data Preparation | Data preparation and transformation    |
|  DAX                                     | Measures and calculated columns        |
|  Power BI Data Model                     | Relationships and analytical structure |
|  Field Parameters                        | Dynamic Measure selection              |
|  Bookmarks                               | Interactive dashboard states           |
|  Drill-through                           | Detailed trip analysis                 |
|  Page Navigation                         | Dashboard navigation                   |
|  Tooltips                                | User guidance                          |
|  Web URL Action                          | GitHub project access                  |

## 3. Business Requirements

The original business requirement is divided into three dashboard sections.

# Dashboard 1 — Overview Analysis

Required KPIs:

- Total Bookings
- Total Booking Value
- Average Booking Value
- Total Trip Distance
- Average Trip Distance
- Average Trip Time

Required analysis:

- Dynamic Measure Selector
- Payment Type analysis
- Day/Night Trip analysis
- Vehicle Type Analysis
- Total Bookings by Day
- Location Analysis
- Top 5 Booking Locations
- Preferred Vehicle by Pickup Location
- Farthest Trip

The requirement also specifies dynamic titles, slicers, tooltips, sorting, filtering, and conditional formatting.

## Dashboard 2 — Time Analysis

Required analysis:

- Dynamic Measure Selector
- Pickup Time by 10-minute intervals
- Day Name analysis
- Hour × Day Heatmap
- Dynamic visual updates based on measure selection

## Dashboard 3 — Details

Required functionality:

- Detailed trip grid
- Drill-through
- Full-data bookmark
- Detailed record exploration

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## 4. Project Development Workflow

The project was developed using the following workflow:

Excel Source Data



Power BI Data Import  
↓  
Data Modeling  
↓  
Relationships  
↓  
Date & Time Transformations  
↓  
DAX Measures  
↓  
Dynamic Measure Parameter  
↓  
Dashboard 1 – Overview Analysis  
↓  
Dashboard 2 – Time Analysis  
↓  
Dashboard 3 – Details  
↓  
Drill-through & Bookmarks  
↓  
Navigation & Interactive Controls  
↓  
Business Insights  
↓  
Business Recommendations

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## 5. Data Sources

Two Excel workbooks were used in the project.

### 5.1 Trip Details

**File:** Uber Trip Details.xlsx

This is the main transactional table containing individual Uber trips.

#### Main Fields

| Field                                                                                               | Description                 |
|-----------------------------------------------------------------------------------------------------|-----------------------------|
|  Trip ID         | Unique trip identifier      |
|  Pickup Time     | Trip pickup date and time   |
|  Drop Off Time   | Trip drop-off date and time |
|  passenger_count | Number of passengers        |
|  trip_distance   | Trip distance               |
|  PULocationID    | Pickup location ID          |
|  DOLocationID    | Drop-off location ID        |
|  fare_amount     | Base fare                   |
|  Surge Fee       | Additional surge fee        |
|  Vehicle         | Vehicle type                |

| Field                                                                                         | Description    |
|-----------------------------------------------------------------------------------------------|----------------|
|  Payment_type | Payment method |

## 5.2 Location Table

**File:** Location Table.xlsx

This is the location lookup table.

| Field                                                                                        | Description                |
|----------------------------------------------------------------------------------------------|----------------------------|
|  LocationID | Unique location identifier |
|  Location   | Location name              |
|  City       | City/area classification   |

## 6. Data Preparation

### 6.1 Pickup Date

The original `Pickup Time` field contains both date and time.

A separate date field was created:

```
Pickup Date =  
DATE(  
    YEAR('Trip Details'[Pickup Time]),  
    MONTH('Trip Details'[Pickup Time]),  
    DAY('Trip Details'[Pickup Time])  
)
```

#### Purpose

- Date filtering
- Daily analysis
- Calendar relationship
- Time-based analysis

### 6.2 Pickup Hour for Time Analysis

A time-only field was created:

```
Pickup Hour (HH MM SS) =  
TIME(  
    HOUR('Trip Details'[Pickup Time]),  
    MINUTE('Trip Details'[Pickup Time]),  
    SECOND('Trip Details'[Pickup Time])  
)
```

This field was later grouped into **10-minute bins**.

## Purpose

- Intraday demand analysis
  - Pickup-time area chart
  - Peak and off-peak analysis
- 

## 6.3 Pickup Hour

A separate hour column was created:

```
Pickup Hour =  
HOUR('Trip Details'[Pickup Time])  
This returns values from 0 to 23.
```

## Purpose

- Hour × Day heatmap
  - Hourly demand analysis
  - Hourly summary bar chart
- 

## 6.4 Day / Night Classification

A calculated column was created:

```
Trip (Day/ Night) =  
VAR HourofDay =  
    HOUR('Trip Details'[Pickup Time])  
RETURN  
IF(  
    HourofDay >= 17 || HourofDay < 6,  
    "Night Trip",  
    "Day Trip"  
)
```

## Classification

| Time                                                                                                | Classification |
|-----------------------------------------------------------------------------------------------------|----------------|
|  6:00 AM–4:59 PM | Day Trip       |
|  5:00 PM–5:59 AM | Night Trip     |

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# 7. Data Modeling

## 7.1 Location Relationships

The Trip Details table was connected to the Location Table using `LocationID`.

## Pickup Relationship

Trip Details[PULocationID]



Location Table[LocationID]

**Cardinality:** Many-to-One ( \*:1 ) **Cross-filter:** Single **Status:**  Active

## Drop-off Relationship

Trip Details[DOLocationID]



Location Table[LocationID]

**Cardinality:** Many-to-One ( \*:1 ) **Cross-filter:** Single **Status:**  Inactive

The inactive relationship is activated when drop-off analysis is required.

---

# 8. Calendar Table

A dedicated Calendar table was created using DAX:

Calendar table =

```
CALENDAR(  
    MIN('Trip Details'[Drop Off Time]),  
    MAX('Trip Details'[Pickup Time])  
)
```

The Calendar table provides a dedicated date dimension for time-based analysis.

---

# 9. Calendar Relationship

The following relationship was created:

Calendar table[Date]



Trip Details[Pickup Date]

## Relationship Settings

- **Cardinality:** One-to-Many
  - **Cross-filter:** Single
  - **Status:**  Active
- 

# 10. Calendar Table Calculated Columns

## 10.1 Day Name

Day Name =

```
FORMAT(  
    'Calendar table'[Date], "ddd")
```

This produces:

Mon

Tue

Wed  
Thu  
Fri  
Sat  
Sun

---

## 10.2 Day Number

```
Day Num =  
WEEKDAY(  
    'Calendar table'[Date],2)
```

### Day Number Mapping

| Day       | Number |
|-----------|--------|
| Monday    | 1      |
| Tuesday   | 2      |
| Wednesday | 3      |
| Thursday  | 4      |
| Friday    | 5      |
| Saturday  | 6      |
| Sunday    | 7      |

The `Day Name` column was sorted using **Sort by Column** → **Day Num** to ensure correct Monday-to-Sunday ordering.

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## 11. DAX KPI Measures

### 11.1 Total Bookings

```
Total Bookings =  
COUNT('Trip Details'[Trip ID])
```

#### Purpose

Counts the total number of trip bookings.

---

### 11.2 Total Booking Value

```
Total Booking value =  
SUM('Trip Details'[fare_amount])  
+ SUM('Trip Details'[Surge Fee])
```

#### Purpose

Calculates the combined base fare and surge fee.

---

## 11.3 Average Booking Value

```
Avg Booking Value =  
DIVIDE(  
    [Total Booking value],  
    [Total Bookings],  
    BLANK()  
)
```

### Purpose

Calculates the average booking value per trip.

---

## 11.4 Total Trip Distance

The KPI display measure was created as:

```
Total Trip Distance =  
VAR TotalMiles =  
    SUM('Trip Details'[trip_distance]) / 1000  
RETURN  
    CONCATENATE(  
        FORMAT(TotalMiles, "0"), "K miles")
```

A numeric measure was also created for dynamic visual analysis:

```
Total Trip Distance Measure =  
SUM('Trip Details'[trip_distance])
```

---

## 11.5 Average Trip Distance

```
Avg Trip Distance =  
VAR AvgMiles =  
    ROUND(  
        AVERAGE('Trip Details'[trip_distance]), 0)  
RETURN  
  
    CONCATENATE(AvgMiles, " miles")
```

---

## 11.6 Average Trip Time

```
Avg Trip Time =  
VAR AvgTime =  
    AVERAGEX(  
        'Trip Details',  
        DATEDIFF(  
            'Trip Details'[Pickup Time],  
            'Trip Details'[Drop Off Time],  
            MINUTE  
        )  
    )  
RETURN  
    CONCATENATE(FORMAT(AvgTime, "0"), "min")
```

---

# 12. Dynamic Measure Parameter

A Field Parameter named **Dynamic Measure** was created.

```
Dynamic Measure = {  
    ("Total Bookings",  
        NAMEOF('Trip Details'[Total Bookings]), 0),  
  
    ("Total Booking value",  
        NAMEOF('Trip Details'[Total Booking value]), 1),  
  
    ("Total Trip Distance",  
        NAMEOF('Trip Details'[Total Trip Distance Measure]), 2)  
}
```

## Available Selections

- Total Bookings
- Total Booking Value
- Total Trip Distance

The parameter is used across the Time Analysis visuals to dynamically switch the selected metric.

---

# 13. Dynamic Titles

A dynamic title was created using the parameter order.

```
Dynamic Title =  
IF(  
    'Dynamic Measure'[Dynamic Measure Order] = 0,  
    "Total Bookings",  
    IF(  
        'Dynamic Measure'[Dynamic Measure Order] = 1,  
        "Total Booking Value",  
        IF(  
            'Dynamic Measure'[Dynamic Measure Order] = 2,  
            "Total Trip Distance",  
            "Other"  
        )  
    )  
)
```

## Pickup Time Title

```
Title for Pickup Time =  
SELECTEDVALUE('Dynamic Measure'[Dynamic Title])  
    & " by Pickup Time"
```

## Day Name Title

```
Title for Day Name =  
SELECTEDVALUE('Dynamic Measure'[Dynamic Title])  
    & " by Day Name"
```

These dynamic titles ensure that visual titles remain synchronized with the selected metric.

# 14. Location Analysis Measures

## 14.1 Most Frequent Pickup Point

Most Frequent Pickup Point =

```
VAR PickPoint =  
    TOPN(  
        1,  
        SUMMARIZE(  
            'Trip Details',  
            'Location Table'[Location],  
            "Pickup Point",  
            COUNT('Trip Details'[Trip ID])  
        ),  
        [Pickup Point],  
        DESC  
    )  
RETURN  
    CONCATENATEX(  
        PickPoint,  
        'Location Table'[Location],  
        ", "  
    )
```

### Result

📍 **Penn Station/Madison Sq West**

---

## 14.2 Most Frequent Drop-off Point

The drop-off measure uses the inactive relationship through `USERELATIONSHIP()`.

Most Frequent Dropoff Point =

```
VAR DropOffCounts =  
    ADDCOLUMNS(  
        SUMMARIZE(  
            'Trip Details',  
            'Location Table'[Location]  
        ),  
        "DropOffCount",  
        CALCULATE(  
            COUNT('Trip Details'[Trip ID]),  
            USERELATIONSHIP(  
                'Trip Details'[DOLocationID],  
                'Location Table'[LocationID]  
            )  
        )  
    )  
VAR RankedDropoffs =  
    ADDCOLUMNS(  
        DropOffCounts,  
        "Rank",  
        RANKX(  
            DropOffCounts,
```

```

        [DropOffCount],
        ,
        DESC,
        DENSE
    )
)

VAR TopDropOff =
    FILTER(
        RankedDropoffs,
        [Rank] = 1
    )

RETURN
    CONCATENATEX(
        TopDropOff,
        'Location Table'[Location],
        ", "
    )

```

## Result

 **Upper East Side North**

---

## 14.3 Farthest Trip

```

Farthest Trip =
VAR MaxDistance =
    MAX('Trip Details'[trip_distance])

VAR PickupLocation =
    LOOKUPVALUE(
        'Location Table'[Location],
        'Location Table'[LocationID],
        CALCULATE(
            SELECTEDVALUE('Trip Details'[PULocationID]),
            'Trip Details'[trip_distance] = MaxDistance
        )
    )

VAR DropoffLocation =
    LOOKUPVALUE(
        'Location Table'[Location],
        'Location Table'[LocationID],
        CALCULATE(
            SELECTEDVALUE('Trip Details'[DOLocationID]),
            'Trip Details'[trip_distance] = MaxDistance
        )
    )

RETURN
    "Pickup: " & PickupLocation &
    " → Drop-off: " & DropoffLocation &
    " (" & FORMAT(MaxDistance, "0.0") & " miles)"

```

## Result

 **Lower East Side → Crown Heights North (144.1 miles)**

# 15. Dashboard Features

The completed report includes:

-  KPI Cards
-  Date Slicers
-  City Slicer
-  Dynamic Measure Selector
-  Dynamic Titles
-  Payment Type Analysis
-  Day/Night Analysis
-  Vehicle Type Analysis
-  Location Analysis
-  Top 5 Location Analysis
-  Time Analysis
-  Heatmap
-  Drill-through
-  Bookmarks
-  Page Navigation
-  Clear All Slicers
-  Tooltips
-  GitHub Repository Link
-  Interactive Icons

## 16. Dashboard 1 — Overview Analysis

### Objective

The Overview dashboard provides a high-level summary of booking performance, revenue, distance, time, vehicle performance, payment behavior, and location demand.

### KPI Cards

| KPI                                                                                                       | Result            |
|-----------------------------------------------------------------------------------------------------------|-------------------|
|  Total Bookings        | <b>104K</b>       |
|  Total Booking Value   | <b>\$1.6M</b>     |
|  Average Booking Value | <b>\$15.0</b>     |
|  Total Trip Distance   | <b>349K miles</b> |
|  Average Trip Distance | <b>3 miles</b>    |
|  Average Trip Time     | <b>16 min</b>     |

### Visualizations

## Payment Type Analysis

Shows booking performance across:

-  Uber Pay
-  Cash
-  Amazon Pay
-  Google Pay

## Day / Night Analysis

Compares:

-  Day Trip
-  Night Trip

## Vehicle Type Analysis

Compares:

-  UberXL
-  UberX
-  Uber Green
-  Uber Comfort
-  Uber Black

using:

- Total Bookings
- Total Booking Value
- Average Booking Value
- Total Trip Distance

## Daily Booking Value

An Area Chart  displays daily booking value across the selected date period.

## Location Analysis

Includes:

- Most Frequent Pickup Point
  - Most Frequent Drop-off Point
  - Farthest Trip
  - Top 5 Booking Locations
  - Most Preferred Vehicle for Pickup Location
- 

# 17. Dashboard 2 — Time Analysis

## Objective

The Time Analysis dashboard focuses on booking behavior across different hours and days.

# Visualizations

## 17.1 Pickup Time — 10-Minute Area Chart

- X-axis: Pickup Time in 10-minute bins
- Y-axis: Dynamic Measure

## 17.2 Day Name — Area Chart

- X-axis: Day Name
- Y-axis: Dynamic Measure
- Sorting: Day Num

## 17.3 Hour × Day Heatmap 🔥

- Rows: Pickup Hour 0–23
- Columns: Monday–Sunday
- Values: Dynamic Measure

## 17.4 Hourly Summary Bar Chart

A horizontal bar chart provides an hourly summary aligned with the heatmap.

## Dynamic Measure Options

All Time Analysis visuals can switch between:

- Total Bookings
- Total Booking Value
- Total Trip Distance

This directly fulfills the dynamic Time Analysis requirement.

---

# 18. Dashboard 3 — Details

## Objective

The Details page provides granular trip-level data.

## Detailed Grid Fields

-  Trip ID
-  Pickup Hour
-  Pickup Date
-  Vehicle
-  Payment Type
-  No. of Passengers
-  Trip Distance
-  Booking Value
-  Pickup Location

-  Total Bookings

## Drill-through

Drill-through functionality was configured using:

- Total Booking Value
- Total Bookings
- Total Trip Distance Measure

This allows users to move from analytical visuals to detailed trip records.

## Full Data Bookmark

A bookmark named:

**Show all unfiltered Data**

was created to return the details view to the complete dataset.

---

# 19. Navigation & Interactive Controls

## Page Navigation

Custom navigation buttons were created for:

- Overview Analysis
- Time Analysis
- Details

## Home Icon

**Action:** Show Overview Dashboard

**Tooltip:**

 Ctrl + Click to View Overview Dashboard

## Information Icon

**Action:** Show All Data Fields Info

**Tooltip:**

 Ctrl + Click to View Raw Data Details

## Restart Icon

**Action:** Clear All Slicers

**Tooltip:**

 Ctrl + Click to Clear All Filters

**GitHub Icon** 

**Action:** Web URL to GitHub Repository

**Tooltip:**

 Ctrl + Click to View Project on GitHub

These controls create a consistent app-like navigation experience.

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## 20. Overview Analysis — Business Insights

### 20.1 Overall Performance

The dashboard records approximately:

-  **104K bookings**
-  **\$1.6M booking value**
-  **\$15 average booking value**
-  **349K miles total trip distance**
-  **3 miles average trip distance**
-  **16 minutes average trip time**

The dataset therefore represents a high-volume trip environment dominated by relatively short urban journeys.

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### 20.2 Payment Type

#### Booking Volume

| Payment Type                                                                                   | Bookings         | Share  |
|------------------------------------------------------------------------------------------------|------------------|--------|
|  Uber Pay   | 70K              | 67.03% |
|  Cash       | 33K              | 32.23% |
|  Amazon Pay | 1K               | 0.56%  |
|  Google Pay | Very small share | —      |

#### Booking Value

| Payment Type                                                                                 | Booking Value | Share  |
|----------------------------------------------------------------------------------------------|---------------|--------|
|  Uber Pay   | \$1.099M      | 70.74% |
|  Cash       | \$443K        | 28.54% |
|  Amazon Pay | \$9K          | 0.55%  |

## Insight

**Uber Pay is the dominant payment channel**, accounting for the majority of both booking volume and booking value.

# 21. Day / Night Insights

## Booking Volume

| Trip Type                                                                                      | Bookings | Share  |
|------------------------------------------------------------------------------------------------|----------|--------|
|  Day Trip     | 68K      | 65.28% |
|  Night Trip | 36K      | 34.72% |

## Booking Value

| Trip Type                                                                                      | Booking Value | Share  |
|------------------------------------------------------------------------------------------------|---------------|--------|
|  Day Trip   | \$975K        | 62.75% |
|  Night Trip | \$579K        | 37.25% |

## Trip Distance

| Trip Type                                                                                      | Distance   |
|------------------------------------------------------------------------------------------------|------------|
|  Day Trip   | 211K miles |
|  Night Trip | 138K miles |

## Insight

Day trips dominate the dataset across **bookings, booking value, and total distance**.

# 22. Vehicle Analysis

# Vehicle Performance

| Vehicle Performance                                                                            |          |               |                   |          |
|------------------------------------------------------------------------------------------------|----------|---------------|-------------------|----------|
| Vehicle                                                                                        | Bookings | Booking Value | Avg Booking Value | Distance |
|  UberXL       | 16,698   | \$249,424     | \$14.9            | 55,721   |
|  UberX        | 38,744   | \$583,880     | \$15.1            | 131,496  |
|  Uber Green   | 14,498   | \$216,181     | \$14.9            | 48,778   |
|  Uber Comfort | 17,078   | \$253,995     | \$14.9            | 56,790   |
|  Uber Black   | 16,710   | \$250,192     | \$15.0            | 56,149   |

## Insight

**UberX is the leading vehicle category**, with the highest booking volume, highest booking value, and highest total trip distance.

# 23. Location Insights

## Most Frequent Pickup Point

 **Penn Station/Madison Sq West**

## Most Frequent Drop-off Point

 **Upper East Side North**

## Farthest Trip

 **Lower East Side → Crown Heights North — 144.1 miles**

## Top 5 Pickup Locations

| Rank                                                                                  | Location                     | Bookings |
|---------------------------------------------------------------------------------------|------------------------------|----------|
|  1 | Penn Station/Madison Sq West | 4,475    |
|  2 | Upper East Side location     | 4,459    |
|  3 | Upper East Side location     | 4,124    |
| 4                                                                                     | Lenox Hill East              | 3,955    |
| 5                                                                                     | Upper West Side              | 3,752    |

## Insight

## 24. Time Analysis — Total Bookings

### Hourly Pattern

Booking activity is relatively low during the early morning.

Demand increases strongly after approximately **6:00 AM**, continues rising through the daytime, reaches its strongest levels during the afternoon, and then decreases through the evening.

### Day-of-Week Pattern

| Day       | Bookings                                                                                         |
|-----------|--------------------------------------------------------------------------------------------------|
| Monday    | 14.7K                                                                                            |
| Tuesday   | 15.1K                                                                                            |
| Wednesday | 15.7K                                                                                            |
| Thursday  | 11.2K                                                                                            |
| Friday    | 9.3K                                                                                             |
| Saturday  | 18.7K                                                                                            |
| Sunday    |  <b>19.2K</b> |

### Insight

**Sunday has the highest booking volume**, followed by Saturday. Friday records the lowest booking volume.

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## 25. Time Analysis — Total Booking Value

### Day-of-Week Booking Value

| Day       | Booking Value                                                                              |
|-----------|--------------------------------------------------------------------------------------------|
| Monday    | \$213K                                                                                     |
| Tuesday   | \$224K                                                                                     |
| Wednesday | \$239K                                                                                     |
| Thursday  | \$173K                                                                                     |
| Friday    | \$146K                                                                                     |
| Saturday  | \$276K                                                                                     |
| Sunday    |  \$283K |

## Insight

Sunday generates the highest booking value, followed by Saturday.

The pickup-time booking-value chart also shows stronger financial activity during the daytime and afternoon period.

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# 26. Time Analysis — Total Trip Distance

## Day-of-Week Distance

| Day       | Distance                                                                                       |
|-----------|------------------------------------------------------------------------------------------------|
| Monday    | 47K                                                                                            |
| Tuesday   | 49K                                                                                            |
| Wednesday | 53K                                                                                            |
| Thursday  | 41K                                                                                            |
| Friday    | 36K                                                                                            |
| Saturday  |  <b>62K</b> |
| Sunday    |  <b>62K</b> |

## Insight

Saturday and Sunday record the highest total travel distance, while Friday records the lowest.

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# 27. Cross-Metric Time Insight

A major observation from the Dynamic Measure analysis is that the three metrics show broadly similar temporal patterns.

## Total Bookings

Higher during daytime and afternoon periods.

## Total Booking Value

Higher during the same high-demand periods.

## Total Trip Distance

Also increases during stronger demand periods.

## Overall Finding

**Periods of higher booking demand generally coincide with higher booking value and greater total trip distance.**

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# 28. Business Recommendations

## 28.1 Driver Allocation 🚦

Increase driver availability during the strongest daytime and afternoon demand periods.

## 28.2 Weekend Capacity Planning 📅

Prioritize Saturday and Sunday because they produce the highest booking volume, booking value, and trip distance.

## 28.3 High-Demand Locations 📍

Focus driver availability around major pickup locations such as:

- Penn Station/Madison Sq West
- Upper East Side
- Lenox Hill East
- Upper West Side

## 28.4 Vehicle Allocation 🚗

Maintain sufficient **UberX** availability because it is the strongest-performing vehicle category.

## 28.5 Payment Strategy 💳

Uber Pay is the dominant payment method, so digital-payment adoption can continue to be supported while maintaining cash availability.

## 28.6 Outlier Trip Monitoring ⚠️

The 144.1-mile trip is significantly above the 3-mile average and may warrant operational or data-quality review.

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## 29. Requirement vs Implementation

| Business Requirement | Implementation       | Status      |
|----------------------|----------------------|-------------|
| KPI Analysis         | KPI Cards            | ✓ Completed |
| Dynamic Measure      | Field Parameter      | ✓ Completed |
| Payment Type         | Donut Chart          | ✓ Completed |
| Day/Night            | Donut Chart          | ✓ Completed |
| Vehicle Analysis     | Matrix               | ✓ Completed |
| Booking Trend        | Area Chart           | ✓ Completed |
| Pickup Location      | DAX Measure          | ✓ Completed |
| Drop-off Location    | USERELATIONSHIP()    | ✓ Completed |
| Farthest Trip        | DAX Measure          | ✓ Completed |
| Top 5 Locations      | Bar Chart            | ✓ Completed |
| Preferred Vehicle    | Bar Chart            | ✓ Completed |
| Pickup Time Analysis | 10-minute Area Chart | ✓ Completed |
| Day Name Analysis    | Area Chart           | ✓ Completed |
| Hour × Day Analysis  | Heatmap              | ✓ Completed |
| Details Grid         | Table                | ✓ Completed |
| Drill-through        | Details Page         | ✓ Completed |
| Full Data Bookmark   | Bookmark             | ✓ Completed |
| Page Navigation      | Navigation Buttons   | ✓ Completed |
| Clear Filters        | Clear All Slicers    | ✓ Completed |
| Data Information     | Bookmark + Info Icon | ✓ Completed |
| GitHub Link          | Web URL Action       | ✓ Completed |

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## 30. Dashboard Limitations

The following limitations should be noted based on the current project implementation:

- ⚠ The analysis is limited to the fields available in the provided Uber dataset.
- ⚠ Customer demographic analysis is not available because demographic attributes are not present.
- ⚠ Profitability analysis is limited because the project does not contain operating-cost information.
- ⚠ The available location data supports location-level trip analysis but does not provide broader regional business attributes.

- ⚠️ The longest-trip result should be treated as an observed record and may require separate validation if used for operational decisions.

## 31. Future Enhancements

Future versions of the dashboard could include:

- 👤 Customer segmentation
- 💰 Customer lifetime value
- 📊 Profitability analysis
- 💵 Cost and margin analysis
- 🚗 Driver performance analysis
- 🔮 Forecasting
- 📈 Demand prediction
- ⚡ Surge pricing analysis
- 🌤️ Weather/event impact analysis
- 📍 Geographic maps
- 🌐 Regional demand forecasting
- 🔄 Real-time or scheduled data refresh
- 📄 Automated reporting

## 32. Final Dashboard Summary

### Dashboard 1 — Overview Analysis

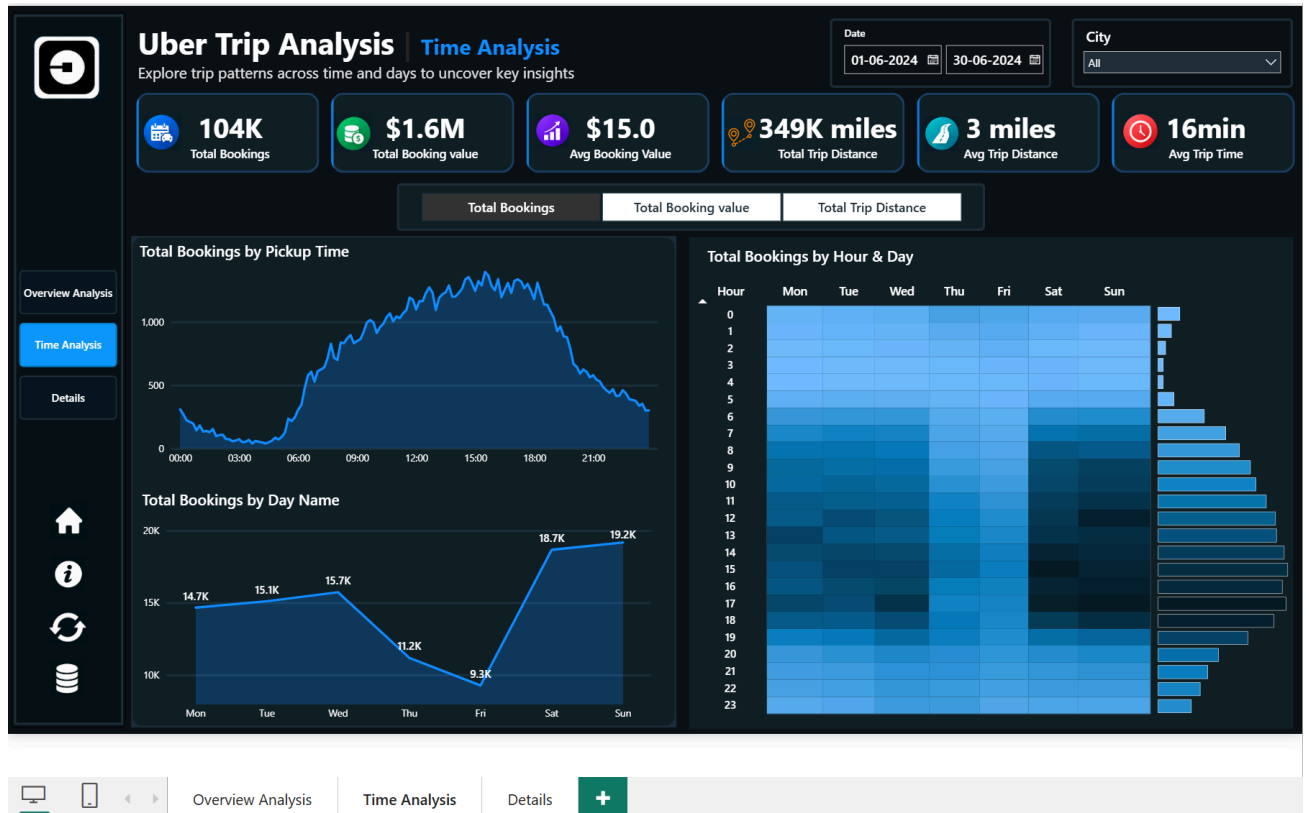


Provides the executive-level view of:

- KPIs

- Payment behavior
- Day/night demand
- Vehicle performance
- Revenue trend
- Location analysis

## Dashboard 2 — Time Analysis



Provides detailed analysis of:

- Pickup time
- Day-of-week demand
- Hour × day demand
- Dynamic booking value
- Dynamic trip distance

## Dashboard 3 — Details



Uber Trip Analysis

Detailed Trip & Booking Records

Details




Overview Analysis

Time Analysis

Details






| Trip ID | Pickup Hour | Pickup Date  | Vehicle      | Payment Type | No of Passengers | Trip Distance | Booking value | Pickup Location              | Total Bookings |
|---------|-------------|--------------|--------------|--------------|------------------|---------------|---------------|------------------------------|----------------|
| 1       | 00:42:50    | 01 June 2024 | UberX        | Uber Pay     | 1                | 6             | \$22          | East Village                 | 1              |
| 10      | 00:54:40    | 01 June 2024 | UberX        | Uber Pay     | 2                | 2             | \$12          | Lenox Hill East              | 1              |
| 11      | 00:18:51    | 01 June 2024 | UberXL       | Uber Pay     | 1                | 7             | \$28          | Clinton East                 | 1              |
| 12      | 00:17:18    | 01 June 2024 | Uber Comfort | Uber Pay     | 2                | 13            | \$37          | Kips Bay                     | 1              |
| 16      | 00:46:55    | 01 June 2024 | Uber Black   | Uber Pay     | 1                | 2             | \$10          | Kips Bay                     | 1              |
| 17      | 00:46:26    | 01 June 2024 | UberXL       | Uber Pay     | 1                | 3             | \$13          | Midtown North                | 1              |
| 18      | 00:14:47    | 01 June 2024 | Uber Comfort | Uber Pay     | 1                | 3             | \$15          | Greenwich Village South      | 1              |
| 19      | 00:42:52    | 01 June 2024 | UberXL       | Uber Pay     | 1                | 6             | \$21          | East Williamsburg            | 1              |
| 20      | 00:20:49    | 01 June 2024 | Uber Green   | Uber Pay     | 1                | 2             | \$9           | Clinton East                 | 1              |
| 22      | 00:04:52    | 01 June 2024 | UberXL       | Uber Pay     | 1                | 6             | \$26          | Penn Station/Madison Sq West | 1              |
| 23      | 00:50:26    | 01 June 2024 | Uber Comfort | Uber Pay     | 1                | 3             | \$11          | Penn Station/Madison Sq West | 1              |
| 24      | 00:03:50    | 01 June 2024 | Uber Comfort | Uber Pay     | 1                | 5             | \$21          | Clinton East                 | 1              |
| 28      | 00:42:11    | 01 June 2024 | UberX        | Uber Pay     | 1                | 8             | \$33          | Kips Bay                     | 1              |
| 29      | 00:23:24    | 01 June 2024 | UberXL       | Uber Pay     | 1                | 7             | \$24          | Kips Bay                     | 1              |
| 30      | 00:20:55    | 01 June 2024 | UberXL       | Uber Pay     | 6                | 4             | \$19          | Lenox Hill East              | 1              |
| 33      | 01:23:40    | 01 June 2024 | UberX        | Uber Pay     | 1                | 8             | \$23          | Manhattan Valley             | 1              |
| 36      | 01:13:35    | 01 June 2024 | UberX        | Uber Pay     | 1                | 5             | \$18          | Midtown Center               | 1              |
| 40      | 01:05:57    | 01 June 2024 | UberX        | Uber Pay     | 1                | 4             | \$17          | Murray Hill                  | 1              |
| 46      | 01:21:41    | 01 June 2024 | Uber Green   | Uber Pay     | 1                | 5             | \$19          | Murray Hill                  | 1              |
| Total   |             |              |              |              | 146478           | 3,48,934      | \$15,53,673   | 1,03,728                     |                |




Overview Analysis

Time Analysis

Details

+

Provides:

- Trip-level records
- Drill-through
- Full-data view
- Detailed filtering

## 33. Final Project Outcome

The Uber Trip Analysis Dashboard successfully transforms raw Uber trip data into an interactive Power BI reporting solution.

The project demonstrates practical capability in:

- Data import
- Data modeling
- Relationships
- Calendar modeling
- DAX
- Calculated columns
- Field Parameters
- Dynamic titles
- Interactive visualizations
- Drill-through
- Bookmarks
- Page navigation
- User-focused dashboard design
- Business analysis
- Insight generation

The final dashboard allows users to move from **high-level KPIs** → **analytical patterns** → **detailed trip records**, providing a complete reporting workflow.

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## 34. Conclusion

The completed Uber Trip Analysis project demonstrates how Power BI can transform transactional trip data into a structured analytical solution.

The three-dashboard design provides:

-  **Overview Analysis** for executive-level performance monitoring
-  **Time Analysis** for understanding demand patterns
-  **Details** for granular trip-level investigation

The integration of DAX measures, dynamic metric selection, relationships, drill-through, bookmarks, navigation, and interactive controls makes the report more than a static collection of visuals—it functions as an interactive business analysis solution.

The project also demonstrates practical skills relevant to Data Analyst roles, particularly in **data modeling, DAX, visualization, dashboard development, and business insight generation**.

In [ ]:

